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Deferred Acceptance Algorithm:

- firms: $\{f_1, \dots, f_m\}$
- workers: $\{w_1, \dots, w_n\}$
- each party has strict preferences over the other
- matching $\mu: W \cup F \rightarrow W \cup F \cup \{\emptyset\}$
- looking for matches which are stable

* DA: finds stable matching from reported prefs
- is it optimal to report truthfully?

* it is not optimal for the side being proposed to

Proof by Contradiction:

Let γ be the current report and γ_1 is w_1 's truthful report. Assume suboptimal to report γ_1 for w_1 . In which case, reporting $\hat{\gamma}_1$ is better for w_1 .

- μ is matching under γ_1 ; $\hat{\mu}$ is matching under $\hat{\gamma}_1$.

Step 1: $B = \{w \in W \mid \hat{\mu}(w) \succ_w \mu(w)\}$

$\hookrightarrow B$ non-empty as $w_1 \in B$

• $|\hat{\mu}(B)| = |B| \Rightarrow$ same no. firms as workers

• $\hat{\mu}(B) = \mu(B) \Rightarrow$ same set of firms matched

Proving: $\hat{\mu}(B) = \mu(B)$

Take a firm $f \in \hat{\mu}(B)$ and let $w = \mu(f)$.
Let $\hat{w} = \hat{\mu}(f)$ and $\hat{f} = \hat{\mu}(w)$. Assume $w \notin B$.
This implies $f \succ_w \hat{f}$. However, $\hat{\mu}$ is stable, so $\hat{w} \succ_{\hat{f}} w$. This brings about a contradiction, implying μ was unstable given preferences $\succ$. So, $f \in \hat{\mu}(B) \Rightarrow \mu(f) \in B$.
Therefore $w \in B$.



Reassignment of firms $\Rightarrow$ make workers better off.

Step 2: Consider DA when run on $\succ$, leading to μ .

- Let w^* be the last worker of set B to be matched

- w^* proposed to $\mu(w^*) = f^*$. Now, $w^* \in B$ and $f^* \in \hat{\mu}(B)$ so $\exists w' = \hat{\mu}(f^*) \in B$.

This means w' proposed to f^* under $\succ$, but was rejected. So, f^* preferred some $\tilde{w} \notin B$ or $\emptyset$. So, $\tilde{w} \succ_{f^*} w'$ or $\emptyset \succ_{f^*} w'$. But, if $\tilde{w} \notin B$, $\tilde{w}$ and f^* form a blocking pair in $\hat{\mu}$. $\times$ due to instability of $\hat{\mu}$. Unstable if $\emptyset \succ_{f^*} w'$ as well.

$f^* = \mu(w^*)$

$w' \rightarrow f^*$ (reject)

$\tilde{w} \succ_{f^*} w'$ s.t. $\tilde{w} \notin B$

$w^* \succ_{f^*} \tilde{w}$

$w^* \succ_{f^*} \tilde{w} \succ_{f^*} w' \Rightarrow f$ would prefer $\tilde{w}$ to new w' . w^* is happy with new match.

f^* is preferred by $\tilde{w}$ more than $\mu(\tilde{w})$ or $\emptyset$

Assumptions

- 1: strict preferences
- 2: disregard other matches
- 3: need preferences over all elements of other set.

Truncated Preferences do not abide by theorems

In residency, match as couples:

- preferences over pairs
- can tweak algorithm to make abide

Interviews:

- look into this